

UX-0003 — Butlerly Information Architecture and Navigation Model

Version 1.0

Purpose

This document defines the Version 1 information architecture, navigation model, content hierarchy, and device-specific adaptation for Butlerly. It translates UX-0001 and UX-0002 into a structure that supports smartphone-first use, native tablet and desktop experiences, local-first operation, optional future cross-device continuity, optional explainable AI, and privacy controls.

1. Information Architecture Goals

The architecture must:

- Reflect user goals rather than database entities.
- Keep the primary navigation small and stable.
- Keep Home, Transactions, Review, Search, and Settings as the stable Finance V1 primary destinations; Capture remains a contextual/global action.
- Preserve the same mental model across smartphone, tablet, and desktop.
- Allow native device layouts without creating three different products.
- Keep privacy and local-data controls understandable. Trusted-device and synchronization controls are deferred until optional cross-device capability is implemented.
- Support offline use and reveal unavailable connected capabilities honestly.
- Separate routine user tasks from advanced administration.
- Scale beyond Version 1 without exposing future complexity prematurely.

2. Governing Principles

2.1 One Product, Three Native Experiences

Butlerly uses one conceptual architecture across all devices, adapted to the strengths of each platform.

Smartphone

Primary role: capture, quick review, search, alerts, and short assistant interactions.

Tablet

Primary role: review, organization, comparison, and focused continuation.

Desktop

Primary role: deep search, bulk import, export, advanced settings, complex correction, and administration.

The names and meaning of core areas remain consistent across devices even when their presentation changes.

2.2 Intent Before Entity

Navigation labels should express what users want to do.

Preferred:

- Review
- Transactions
- Search
- Assistant (optional, non-primary in Finance V1)
- My Devices (future optional cross-device capability)

Avoid as primary navigation:

- Internal database or ledger terminology
- Document objects
- AI engine
- Sync jobs
- Metadata

Technical entities remain available in detail views and advanced settings where needed.

2.3 Progressive Disclosure

The architecture presents the smallest useful structure first. Advanced filters, provider settings, version history, and administrative controls appear only when relevant.

2.4 Global Actions Are Not Destinations

Capture is available contextually or globally. Search is a primary Finance V1 destination and may also have contextual shortcuts.

2.5 Privacy Is a First-Class Destination

Privacy and Data must be reachable through Settings and contextually from any feature that requests external processing, synchronization, or connected-source access.

3. Version 1 Top-Level Architecture

The recommended top-level areas are:

1. Home
2. Transactions
3. Review
4. Search

5. Settings

Global capabilities:

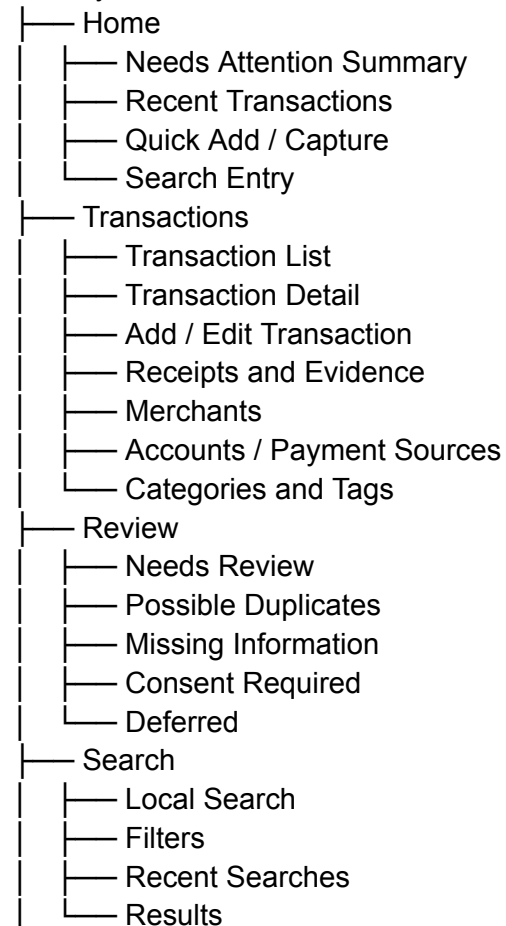
- Capture
- Search
- Notifications and activity status
- Contextual help and explanation

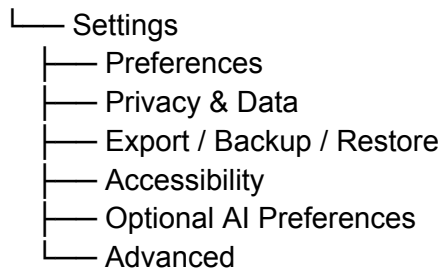
Supporting destinations:

- Record Detail
- Import Center
- Review Session
- My Devices (future optional cross-device capability)
- Privacy & Data
- Connections
- Recently Deleted
- Version History

4. Site Map

Butlerly





Supporting capabilities:

- Capture / Import — contextual actions, not a permanent top-level destination.
- Assistant (optional, non-primary in Finance V1) — optional future capability; not a Finance V1 primary destination.
- My Devices (future optional cross-device capability) / Synchronization — future optional cross-device capability; not required for the Finance V1 MVP.

5. Top-Level Area Definitions

5.1 Home

Purpose

Provide a calm, useful starting point that answers:

- What happened recently?
- What needs my attention?
- What should I do next?

Home is not a dashboard of every available metric. It should prioritize action and continuity.

Required modules:

- Continue where you left off.
- Needs attention count and highest-priority item.
- Recent records.
- Capture action.
- Search entry.
- Sync or processing status only when noteworthy.

Avoid:

- Decorative charts.
- Large balances without context.
- Marketing content.
- Duplicate summaries available elsewhere.

5.2 Review

Purpose

Provide one trusted queue for items that genuinely require user attention.

Review groups work by reason and consequence rather than by internal source.

Primary categories:

- Needs Review
- Possible Duplicate
- Missing Information
- Consent Required
- Import or Connection Issue
- Deferred

Each item must expose:

- What Butlerly believes.
- Why attention is needed.
- Recommended action.
- Alternative actions.
- Deferral option when appropriate.

Review should never become a general notification center.

5.3 Transactions

Purpose

Provide the canonical browsable history of user-controlled financial information.

Records includes multiple views of the same underlying information. The default view should be All Records or Recent Records, not an accounting-specific structure.

Record types:

- Transaction
- Receipt
- Statement or imported document
- Merchant
- Account or source
- Category or label

Relationships:

- Receipt linked to transaction.
- Transaction linked to merchant and account.
- Document linked to imported records.
- Record linked to change history.
- Record linked to AI explanation and source evidence.

Records must support filters without forcing users to build a folder hierarchy.

5.4 Optional Assistant (Future Capability)

Purpose

Allow users to ask questions, understand records, and initiate supported actions using natural language.

Assistant is embedded in the product, not a replacement for navigation.

Assistant modes:

- Global question.
- Record-context question.
- Search assistance.
- Explanation of Butlerly decisions.
- Supported action initiation.

Assistant must preserve context and disclose when external processing is required.

5.5 Settings

Purpose

Provide user control over preferences, privacy, data, devices, services, notifications, and advanced behavior.

Settings is divided into:

- Everyday preferences.
- Privacy and control.
- Connected capabilities.
- Advanced technical controls.

The most sensitive controls must be understandable without technical expertise.

6. Global Capabilities

6.1 Capture

Capture is the highest-priority global action on smartphone and remains readily available on tablet and desktop.

Entry methods:

- Camera.
- Photo library.

- File picker.
- Drag and drop.
- Manual entry.
- Share sheet.

The action opens into the appropriate native capture experience, then returns the user to the originating context or completion state.

6.2 Search

Search is globally available and supports:

- Natural-language queries.
- Structured filters.
- Recent queries.
- Suggested refinements.
- Result previews.
- Direct correction from results.

Search should preserve user filters and query context when opening and returning from a result.

6.3 Activity and Status

Core Finance V1 does not require synchronization. If optional synchronization is introduced later, successful background synchronization should remain quiet.

Status becomes visible when:

- Processing is active long enough to matter.
- A future optional synchronization feature is delayed or failed.
- A device is offline.
- An import completed with exceptions.
- External consent is required.

7. Smartphone Navigation Model

Primary pattern

Bottom navigation with five Finance V1 destinations: Home, Transactions, Review, Search, and Settings. Capture may remain a prominent contextual action.

Recommended tabs:

- Home
- Review
- Transactions
- Assistant (optional, non-primary in Finance V1)

Settings access:

Settings or profile/preferences control in the top app bar where appropriate. A Butlerly account is not required for core operation.

Search:

Persistent search affordance on Home and Records, with a global shortcut in the top app bar.

Capture:

Prominent floating, center-tab, or platform-native primary action.

Smartphone rules:

- Optimize for one-handed use.
- Use focused single-column flows.
- Review one exception at a time by default.
- Keep detail pages scrollable and sectional.
- Preserve bottom navigation when it does not interfere with focused tasks.
- Hide navigation during camera capture and other immersive actions.

8. Tablet Navigation Model

Primary pattern

Sidebar or split-view navigation.

Recommended sidebar:

- Home
- Review
- Transactions
- Assistant (optional, non-primary in Finance V1)
- Settings

Capture and Search remain global actions in the toolbar.

Tablet rules:

- Use list-detail split views for Review, Transactions, Search, and future Devices where enabled.
- Allow side-by-side receipt and transaction comparison.
- Preserve context when opening detail.
- Support keyboard and pointer without compromising touch.
- Adapt between compact and expanded layouts based on window size.

9. Desktop Navigation Model

Primary pattern

Persistent left sidebar with global toolbar and optional secondary pane.

Recommended sidebar:

- Home
- Review
- Transactions
- Assistant (optional, non-primary in Finance V1)
- Settings

Secondary navigation appears contextually:

- Record types and saved views under Transactions.
- Review categories under Review.
- Settings sections under Settings.

Desktop rules:

- Support keyboard-first navigation.
- Provide list-detail and multi-pane workflows.
- Allow bulk selection for reversible actions.
- Support drag and drop.
- Preserve window and panel state where appropriate.
- Make import, export, and advanced filters more discoverable than on smaller devices.

10. Future Optional Cross-Device Continuity in Navigation

This section is a future-design reference and is not part of the Finance V1 MVP. If cross-device continuity is later implemented, it should appear in normal product language.

Examples:

- Continue reviewing 4 items.
- Receipt captured on iPhone is ready.
- Import completed on Mac.
- Changes are available on this device.

Home may show a Continue card when recent context exists.

My Devices is the authoritative place for:

- Device trust.
- Add or remove device.
- Last synchronization.
- Pending changes.
- Manual encrypted transfer.
- Apple synchronization status.

Navigation must not expose provider-specific infrastructure as the primary concept. The user sees devices and availability; provider details are secondary.

11. Record Detail Architecture

Every record detail uses a consistent structure:

1. Identity

- Merchant or source.
- Amount or primary value.
- Date.
- Record type and status.

2. Butlerly Understanding

- Category.
- Linked records.
- Confidence.
- Brief explanation.

3. Source Evidence

- Original receipt, file, or imported source.
- Extracted fields.

4. User Notes and Labels

5. Related Records

6. Activity and Version History

7. Actions

- Edit.
- Ask Butlerly.
- Link or unlink.
- Export or share.
- Delete.

Complex sections remain collapsed until needed.

12. Search Result Architecture

Each result includes:

- Primary identity.
- Amount or key value.
- Date.
- Record type.
- Match reason.
- Relevant snippet or image preview.

- Status when attention is required.

Result groups may include:

- Best matches.
- Transactions.
- Receipts.
- Documents.
- Merchants.

The system should not split results into so many groups that users must understand the data model.

13. Inbox Information Hierarchy

Priority order:

1. Security, privacy, or data-loss risk.
2. Time-sensitive consent or connection issue.
3. Same-record synchronization conflict that could not be resolved automatically.
4. High-consequence uncertain interpretation.
5. Possible duplicate.
6. Missing information.
7. Low-consequence cleanup.

Within an item:

1. Problem statement.
2. Butlerly recommendation.
3. Evidence and reason.
4. Primary decision.
5. Alternatives.
6. Defer or inspect detail.

14. Settings Architecture

14.1 Profile and Preferences

- Locale.
- Currency.
- Date format.
- Appearance.
- Default views.

14.2 Privacy & Data

- Where data is stored.
- What is synchronized.
- Which external providers receive data.

- Export.
- Delete selected data.
- Erase all data.
- Recently Deleted.

14.3 My Devices

- Trusted device list.
- Add device.
- Revoke device.
- Sync status.
- Pending changes.
- AirDrop or encrypted manual transfer.

14.4 Connections

- Financial sources.
- Cloud drives.
- Email or notification sources when supported.
- Permission status.
- Last successful access.

14.5 AI Preferences

- Local AI availability.
- External provider selection.
- Consent behavior.
- Learning from corrections.
- Reset learned preferences.

14.6 Advanced

- Storage diagnostics.
- Import logs.
- Data model or schema information only when useful.
- Developer or troubleshooting controls.

15. Permissions and Consent Entry Points

Permissions must be requested contextually.

Camera

Requested when the user chooses receipt capture.

Files or Photos

Requested when the user chooses import.

Notifications

Requested after Butlerly can explain what notifications will be used for.

Synchronization

Deferred beyond the Finance V1 MVP. If later implemented, request permission only when the user explicitly enables cross-device continuity.

External AI

Requested before the first relevant transfer, with a clear explanation of what data will leave the device.

Every denied permission must have a non-blocking recovery path.

16. Empty States

Home

“Capture or import your first record.”

Inbox

“You’re all caught up.” Include no engagement prompt.

Records

“No records yet.” Offer Capture, Import, or Sample Data.

Search

“Search by merchant, date, amount, category, or a question.”

Assistant

“Ask about your records or how Butlerly organized them.”

My Devices

“This is your first trusted device.” Offer Add Device and explain available methods.

Empty states should teach the next useful action without creating pressure.

17. Offline and Degraded States

Offline mode:

- Core local destinations remain available.
- Connected-only actions are clearly marked.
- Pending changes are shown in status detail, not as persistent alarm banners.
- Search operates over local data.
- Assistant (optional, non-primary in Finance V1) uses local capability when available and explains limitations.

Future optional degraded synchronization:

- Do not block local editing.
- Explain which devices may be out of date.
- Provide retry and manual transfer paths.

18. Navigation State Preservation

The application should preserve:

- Current tab or destination.
- Search query and filters.
- Review position and unresolved item.
- Selected record in split view.
- Draft manual entry.
- Recent review progress.

Future optional behavior: if cross-device continuity is later implemented, Butlerly may offer to continue recent work without forcing identical screen state across form factors.

19. Deep Links and External Entry

Supported external entry points should map to stable destinations:

- Share a receipt → Capture processing.
- Notification about exception → Specific Review item.
- Device pairing link or QR → Add Device.
- Search shortcut → Search.
- File open with Butlerly → Import preview.

After completion, the user returns to a clear product destination rather than an isolated modal.

20. Terminology Standard

Use:

- Home
- Review
- Transactions
- Assistant (optional, non-primary in Finance V1)
- Settings
- Capture
- Search
- My Devices (future optional cross-device capability)
- Privacy & Data
- Needs Review
- Recently Deleted
- Version History

Avoid in primary UI:

- Repository
- Ledger
- Entity
- Classification engine
- Replication
- Conflict resolver
- Object store
- Sync job

21. Version 1 Scope Boundaries

Included:

- Personal financial records.
- Receipts, transactions, and imported documents.
- Search and retrieval.
- AI explanations.
- Review queue.
- Single-device local continuity across sessions; cross-device continuity is deferred beyond the MVP.
- Local and optional external processing controls.

Excluded from primary architecture:

- Full accounting modules.
- Tax preparation.
- Investments.
- Lending.
- Business collaboration.
- Password management.
- Broad Life OS modules.

The architecture should allow future extensions without reserving empty navigation for them in Version 1.

22. Dependencies and Governance

UX-0003 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- Product MVP scope and personas.
- Architecture decisions for identity, local storage, encryption, synchronization, and AI providers.

UX-0003 governs:

- UX-0004 — Core User Flows.
- UX-0005 — Screen and State Inventory.
- UX-0007 — Design System Foundations.
- UX-0008 — Cross-Device Experience and Continuity.
- UX-0010 — Low-Fidelity Wireframe Specification.
- Engineering routing, navigation, and deep-link implementation.

23. Acceptance Criteria

This document is complete when:

- Every Version 1 journey has a clear entry point and destination.
- The top-level navigation contains no unnecessary future modules.
- Smartphone, tablet, and desktop share one mental model with native layouts.
- Home, Transactions, Review, Search, and Settings are consistently defined as the Finance V1 primary destinations; Assistant remains optional and non-primary.
- Privacy and local-data controls are discoverable; future synchronization and trusted-device controls are explicitly non-MVP.
- Offline, empty, degraded, and recovery states are represented.
- Later user flows and wireframes can be derived without inventing new major destinations.

24. BDS-0000 Conformance

BDS-0000 — Butler Design System v1.0 governs visual tokens, concrete theme values, typography, spacing/radius, responsive breakpoints, component/state contracts, accessibility implementation, localization/original-data presentation, and Flutter UI implementation. This information-architecture document governs destinations and hierarchy, but must not override BDS implementation rules.

For Finance V1, the stable visible primary destinations are Home, Transactions, Review, Search, and Settings; Capture is a global/contextual action and Assistant is optional/non-primary. Any older explanatory references to Inbox, Records, or Assistant as primary tabs/sidebars are legacy terminology and are superseded by the Finance V1 destination set above. Internal concepts such as review queues, record browsers, and legacy screen IDs may remain for traceability without becoming primary navigation labels. Core operation requires no Butlerly account. Device trust and synchronization are future/optional capability-driven areas unless product scope explicitly enables them. UI-authored text follows the selected application language; user-entered/imported/scanned/source data remains in its original language by default, and original currency/amount remains authoritative with any reference-currency conversion shown separately.